

LAB 4

Search System, AJAX, and Advanced Features

Lab Title

Dynamic Search and Interactive Features using AJAX

1. Lab Overview

In this lab, students will implement **dynamic search functionality** and advanced features required for the semester project.

Students will integrate:

- AJAX search
- sorting
- pagination or lazy loading
- breadcrumb navigation

2. Learning Objectives

Students will learn to:

- implement AJAX requests
- build search queries
- update content dynamically without page reload
- implement pagination or lazy loading

3. Prerequisites

Students must complete:

- Lab 1
- Lab 2

4. Lab Tasks

Task 1 — Create Search Interface

Create page: [search.php](#)

Add search input:

```
<input type="text" id="searchBox">
```

Task 2 — Implement AJAX Request

Create JavaScript function:

```
function searchItems(){
  let keyword=document.getElementById("searchBox").value;
  fetch("search_items.php?q="+keyword)
  .then(response=>response.text())
  .then(data=>{
    document.getElementById("results").innerHTML=data;
  });
}
```

Task 3 — Create Search API

Create file: `search_items.php`

Example SQL:

```
SELECT * FROM items
WHERE name LIKE '%keyword%'
```

Return matching results.

Task 4 — Implement Sorting

Allow sorting by:

- name
- price

Example SQL:

```
ORDER BY price ASC
```

Task 5 — Implement Pagination or Lazy Loading

Students must implement **one of the following**:

Option A: **Pagination**

Option B: **Lazy loading**

Task 6 — Add Breadcrumb Navigation

Example:

[Home](#) > [Items](#) > [Category](#) > [Item](#)

5. Expected Results

Students should have:

- dynamic search using AJAX
- search results without page reload
- sorting functionality
- pagination or lazy loading
- breadcrumb navigation

6. Deliverables

Students demonstrate:

- working search system
- AJAX requests
- sorting
- pagination or lazy loading